

Referee Report

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1 Summary

This paper develops a three-period model of a continuum of small open economies featuring two frictions: dollar-denominated corporate debt, which generates balance sheet effects through exchange rate movements, and sticky wages, which make monetary policy non-neutral. The presence of dollar debt means that when the domestic currency depreciates, the real repayment burden on entrepreneurs indebted in dollars rises, weakening their net worth and reducing their capacity to invest through a collateral constraint. The presence of sticky wages means that when the central bank raises its policy rate to appreciate the currency and alleviate this burden, the resulting fall in demand for non-tradable domestic goods translates directly into involuntary unemployment, since wages cannot adjust downward. Together, these two frictions create a trade-off for central banks in emerging market economies: stabilising balance sheets requires appreciating the currency, but doing so comes at the cost of employment.

The model builds on [Bianchi and Lorenzoni \(2022\)](#) for the small open economy structure and on [Gabaix and Maggiori \(2015\)](#) for the modelling of frictional global financial intermediation, where the size of aggregate capital flows determines the premium on the dollar interest rate.

2 Paper's Results

The paper's first result (Proposition 1) establishes that when dollar-denominated debt exceeds a certain threshold, a central bank using only conventional monetary policy cannot achieve full employment. It must raise rates above the natural rate (incurring unemployment) in order to appreciate the currency (via CIP logic) to protect entrepreneurs' balance sheets from the burden of higher dollar repayments. The optimal domestic rate is increasing in the level of dollar debt and the US interest rate. This generates a global financial cycle, as central banks in all EMEs simultaneously hike in response to a Fed hike. The paper's main contribution is the bottleneck externality (Proposition 2). It states that when all EMEs tighten simultaneously, each attracts capital flows to appreciate its currency. These flows are processed through the same global intermediary, which charges a higher premium as the volume of intermediation increases. This raises the dollar rate faced

in practice by each EME, which depreciates their currency further and prompts another round of tightening by the EME's central bank. This in turn leaves unemployment and output inefficiently low. Two conditions are required for this externality: non-separable preferences ($\rho > 1$), and frictional intermediation ($\Gamma > 0$).

On the policy side, the paper shows that macroprudential policy (an ex-ante taxation of dollar debt issuance) has positive spillovers (Proposition 6) because it reduces the need for the domestic central bank to tighten in response to a Fed hike (balance sheets are less at risk), which eases the congestion charge generated by the global intermediary. FX interventions (selling dollar reserves to appreciate the peso) appear attractive because they intervene in FX markets without incurring the domestic cost of raising rates. The paper however shows that this generates a moral hazard: entrepreneurs anticipate that the central bank will defend the exchange rate, which makes dollar debt cheaper to repay in the future, encouraging even more issuance in dollars before the Fed even tightens.

3 Literature Review

The starting point of this paper, and this strand of literature, is the empirical observation made by [Rey \(2015\)](#) that capital flows, credit growth, asset prices and monetary policy across countries co-move and are driven by US monetary policy and global risk aversion. Rey coined the term “Global Financial Cycle” (GFC) and argued that it poses a challenge for EMEs: with flexible exchange rates, countries are still unable to isolate themselves from US monetary policy and financial conditions. Subsequent research, such as [Miranda-Agrippino and Nenova \(2022\)](#) or [Cristi et al. \(2024\)](#), amongst others, further refines these findings by analysing propagation and transmission mechanisms. While an active research question in this framework concerns why the US dollar occupies this special position, the author takes this fact as given and explores its consequences.

Additionally, the paper focuses on foreign-currency debt held by private companies on their balance sheets, and its adverse consequences. [Caballero and Krishnamurthy \(2003\)](#) document excessive dollar debt in emerging economies, while [Calvo and Reinhart \(2002\)](#) establishes “fear-of-floating” for emerging market central banks, as they use monetary tools to control their exchange rates to protect balance sheets. More recently, [Ottonello \(2021\)](#) shows how monetary policy becomes contractionary during crises when there is foreign-currency debt, as central banks tighten to mitigate balance sheet effects from exchange rate depreciations.

The model framework of a small open economy is built on the established framework by [Bianchi and Lorenzoni \(2022\)](#), but endogenizes “fear-of-floating” through the issuance of dollar-denominated corporate debt. The financial friction on entrepreneurs follows [Tirole \(2010\)](#) and [Gertler and Rogoff \(1990\)](#), where net worth determines borrowing capacity through a leverage multiplier, and a reduction in net worth leads to amplified declines in investment.

The intermediation friction, where the dollar interest rate faced by EMEs differs from the Fed rate by a premium proportional to aggregate capital flows, draws on [Gabaix and Maggiori \(2015\)](#)'s paper on international liquidity and exchange rate dynamics with financially constrained intermediaries. Moreover, the author builds on five different micro-

foundations that explain this friction: quadratic costs (Bianchi and Lorenzoni, 2022), agency frictions (Gabaix and Maggiori, 2015), position limits with heterogeneous participation costs (Fanelli and Straub, 2021), VaR constraints (Coimbra and Rey, 2024), and risk-averse intermediaries exposed to noise-trader risk (Itskhoki and Mukhin, 2023).

Furthermore, a core section of the paper examines international monetary spillovers and policy coordination, originating with Obstfeld and Rogoff (2002) and Devereux and Engel (2003). Korinek (2017) gives the general framework for assessing when spillovers generate welfare loss, and cooperation results in welfare gains. In the author’s model, the violation stems from the use of a single instrument (monetary policy) to control both employment and the exchange rate. The closest comparison for the spillover mechanism is Fornaro and Romei (2022), who study monetary policy during unbalanced global recoveries. In their model, spillovers arise because all countries seek to run trade deficits to sustain tradable consumption, but since trade must be balanced, the world interest rate rises (as opposed to the present paper, where spillovers arise through the financial intermediary channel).

Caldara et al. (2024) is the most closely related paper in terms of mechanism: they build a model where synchronous monetary tightening has spillover effects because it tightens global financial intermediaries’ constraints, and they document empirically that synchronous tightening episodes are associated with larger effects on output. However, in the present paper, the mechanisms differ in important ways.

Lastly, the paper’s normative results connect to a large literature on macroprudential regulation in open economies. Farhi and Werning (2016) provided the foundational result that pecuniary externalities in borrowing create inefficiency when combined with nominal rigidities, justifying macroprudential policy to correct aggregate demand externalities. The inefficiency in Farhi and Werning (2016) arises from incomplete markets in a domestic context, whereas the author’s bottleneck externality arises specifically from the interaction of dollar debt, frictional intermediation, and simultaneous policy responses across countries.

Recently, Bianchi (2011), Bianchi and Mendoza (2018), and Jeanne and Korinek (2019) study macroprudential policy in small open economies with pecuniary externalities, showing that private agents overborrow relative to the social optimum, an argument extended by the author. Additionally, Basu et al. (2025) and Itskhoki and Mukhin (2023) study optimal policy in a single small open economy with frictional intermediaries and show that FX interventions and macroprudential measures can improve welfare. The author extends these results, from single economies, to the coordination across a group of EMEs.

4 Comments

4.1 Non-segmented intermediation and the endogeneity of the friction parameter

The bottleneck externality relies entirely on the assumption that capital flows from all EMEs pass through a single, non-segmented intermediary pool (equation 17), such that a dollar flowing to Brazil is perfectly substitutable with one flowing to Indonesia from

the intermediary’s perspective. In reality, intermediation is at least partially segmented: different intermediaries specialise in different regions, currencies, and asset classes, and a shock to flows from one EME need not raise the cost of intermediating flows to another by the full amount the model predicts. The author discusses this briefly in Section 4.5 and cites [An and Huber \(2025\)](#) on cross-elasticities across G10 currencies and [Cerutti and Zhou \(2024\)](#) on CIP deviations across EMEs. However, given that non-segmentation is a precondition of any cross-country spillover, rather than a mere simplification, further investigation is required. The paper would benefit substantially from a more explicit treatment of how partial segmentation would affect the bottleneck externality. We would suggest investigating what degree of cross-country substitutability in intermediary flows is needed for the externality to be quantitatively meaningful.

Endogenous Γ and the relevance of micro-foundations. A related concern is the paper’s claim in Appendix A.14 that the specific micro-foundation for the intermediation friction Γ is immaterial, since five different formulations yield the same reduced form. This equivalence holds only when Γ is treated as an exogenous parameter. In the [Itskhoki and Mukhin \(2023\)](#) formulation, $\Gamma = \omega\sigma_\varepsilon^2$, where σ_ε^2 reflects the risk intermediaries face from exchange rate volatility. If “fear of floating” suppresses exchange rate volatility, then Γ falls precisely when central banks are defending their currencies most aggressively. This creates a dampening feedback that runs counter to the bottleneck externality: in the uncoordinated Nash equilibrium, aggressive currency defence lowers Γ and mitigates spillovers, while in the coordinated equilibrium, less intervention raises exchange rate volatility, increases Γ , and partially offsets the coordination gains. The gap between coordinated and uncoordinated outcomes could therefore be substantially smaller than Proposition 7 suggests or even completely neutralised. Appendix D considers Γ increasing with Fed tightening, but does not consider this opposing channel. The case where the sign of the endogeneity of Γ with respect to the policy environment differs across micro-foundations has to be further investigated.

4.2 Absence of a quantitative assessment

The paper is theoretical and deliberately keeps the analysis tractable in order to isolate a general mechanism that is consistent with patterns observed in the data. But without a quantitative exercise, it is difficult to assess whether the bottleneck externality is large enough to be a meaningful driver of the Global Financial Cycle in practice. This concern connects to a technical point: Proposition 7, which characterises the welfare gains from coordination, is derived to first order in Γ , which is valid when the intermediation friction is small (see appendix A.16). The paper does not discuss whether this approximation remains reliable for empirically relevant values of Γ , yet the economic relevance of the bottleneck externality depends on Γ being large enough. A numerical exercise would help assess whether the linear approximation is a reasonable guide for the magnitudes that matter in practice.

The static three-period structure makes it hard to map the mechanisms to business-cycle frequencies or to assess their persistence. It also limits the analysis of moral hazard: in a dynamic setting, entrepreneurs who anticipate repeated central bank interventions (or non-intervention) would have even stronger (weaker) incentives to issue in dollars, a

feedback loop that the three-period model cannot fully capture. A dynamic extension, or at the very least a discussion of how one could embed the key trade-offs in a richer quantitative framework, would help the reader gauge whether the forces identified here are of major importance in a more realistic setting.

Another issue is that the paper characterises the macroprudential trade-off analytically (Proposition 5), but only for $\rho = 1$, precisely the case where the bottleneck externality is absent. While the paper cites [Bergant et al. \(2024\)](#) as empirical support, this is a qualitative consistency argument: the model is not calibrated to match these findings, so it remains unclear how the costs in productive investment compare to the benefits of reducing the congestion externality, especially when $\rho > 1$.

5 Paper's Strengths

A notable strength of the paper is the appendix. The main text presents the simplest possible version of the model (extreme currency mismatch, single tradeable good at a fixed world price, fully rigid wages, symmetric EMEs) to allow the bottleneck externality to emerge with full clarity. The appendices then systematically relax these assumptions and show that the core results survive. Appendix B.1. generalizes to partial currency mismatch, where entrepreneurs earn some dollar revenues alongside their peso revenues. Here, the balance sheet effect is weakened but not fully eliminated. Appendix B.3. replaces rigid wages with flexible wages, removing the aggregate demand channel entirely, which shows that a similar tradeoff arises for the central bank (through the investment channel instead). Appendix E allows for tradeable price inflation, which dampens the policy spillovers but does not eliminate them. Finally, Appendix F addresses the empirical finding in [De Leo et al. \(2024\)](#) that EMEs sometimes ease rather than tighten in response to Fed hikes, and shows that the bottleneck externality persists because EME rates are still higher than what generates full employment. The thoroughness of the appendix has its merit, and convinces the reader that the mechanism survives each generalization. However, since each extension dampens the magnitude of the bottleneck externality, one is left wondering how much of this channel persists when several of these real-world complications are present simultaneously. This, in turn, is a question that the paper's purely analytical approach cannot answer and would surely benefit from a quantitative exercise.

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